README

Rule-Based AI Chatbot | Project 1

Project Overview

This project is a simple rule-based chatbot written in Python. It responds to a small set of predefined messages using **if-elif-else** conditions and continues chatting in a loop until the user types an exit command.

Features

Feature	Description
Greeting responses	Responds to hello, hi, and hey.
Predefined answers	Answers simple questions such as “how are you” and “what is your name”.
Continuous conversation	Uses a while loop to keep the chatbot running.
Exit commands	Stops when the user types bye, exit, or quit.
Fallback response	Replies when it does not recognize the input.

Requirements

• Python 3.10 or newer is recommended. The project uses only Python's built-in functions and does not require external packages.

How to Run

1. Save the chatbot code in a file named **project 1.py** (or another filename ending in .py).

2. Open VS Code and open the folder containing the file.

3. Open the integrated terminal: Terminal → New Terminal.

4. Run the script using the Python launcher:

```
py -3.12 "C:\Users\YourName\project 1.py"
```

Replace the example path with the actual path to your Python file. If the py -3.12 command is not available, try **py --version** to check the installed Python version, then use the matching launcher command.

Example Conversation

```
Welcome to AI Chatbot!
Bot: What is your name?
YourName
Bot: Nice to meet you, YourName!
You: hello
Bot: Hello! How can I help you?
You: bye
Bot: Goodbye! Have a nice day.
```

How It Works

The program reads the user's message, converts it to lowercase, and removes extra spaces. It then checks the message against predefined rules using if-elif-else statements. The while loop repeats the process until an exit command is received.

Limitations

This is a rule-based chatbot, not a machine-learning model. It does not learn from conversations and can only respond to inputs covered by its rules. New responses can be added by extending the conditional statements.

Troubleshooting

• **Python is blocked or the wrong version runs:** Check which Python version is available with **py --version**. Use an approved Python installation and follow your computer administrator's security policy.

• **The script runs but prints nothing:** Save the file with Ctrl+S, verify that it contains the chatbot code, and run the correct file path.

• **File not found:** Confirm the filename and folder. Use quotes around paths containing spaces.